ثانوية المدينة التعليمية

Education City High School

2.8

Inverse Functions

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: ______________________ Class: ____________ Date: ____________

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: A function has an inverse function exactly when it is one-to-one on its domain.

2. Write one precise sentence explaining this principle: The horizontal line test checks one-to-one behavior graphically.

3. Write one precise sentence explaining this principle: Inverse graphs reflect across $y = x$.

4. Write one precise sentence explaining this principle: Domains and ranges swap; restrictions may be needed.

Core

5. Find the inverse of $f(x) = 3x - 7$.

6. Restrict $f(x) = x^2$ so its inverse is a function and find the inverse.

7. Why is $1/f(x)$ not inverse notation?

8. State what the formula $f^{-1}(f(x)) = x$ tells you and name any required restriction.

9. State what the formula $f(f^{-1}(x)) = x$ tells you and name any required restriction.

10. State what the formula $(a,b) \in f \iff (b,a) \in f^{-1}$ tells you and name any required restriction.

Medium

11. Find the inverse and state its domain: $f(x) = 5x - 4$.

12. Find the inverse and state its domain: $f(x) = (x-2)^3 + 1$.

13. Find the inverse and state its domain: $f(x) = \frac{x+1}{x-3}$.

14. Find the inverse and state its domain: $f(x) = x^2 - 4,\ x \geq 0$.

15. Find the inverse and state its domain: $f(x) = 2^x$.

Hard

16. Determine whether $f(x) = x^3 - 3x$ is one-to-one on all real numbers.

17. Restrict $f(x) = (x-3)^2 + 2$ to an interval where it is one-to-one and find inverse.

18. If f has domain [1,7] and range [-2,5], state inverse domain/range.

19. Verify algebraically that $f(x) = \frac{x-1}{x+2}$ and $g(x) = \frac{-2x-1}{x-1}$ are inverses.

Difficult

20. Find all linear functions that are their own inverses.

21. Can a non-one-to-one function have an inverse relation? Explain.

22. Find inverse of $f(x) = ab^{x-h} + k$, assuming a nonzero.

Challenge / HOT

23. Prove the inverse of a composition reverses order.

24. Construct a function with domain all real numbers whose inverse has domain (0,infinity).

AP-Style Multiple Choice

25. Inverse graph is reflection across
- **(A)** x-axis
- **(B)** y-axis
- **(C)** y=x
- **(D)** origin

26. Functions have inverse functions when they are
- **(A)** even
- **(B)** one-to-one
- **(C)** periodic

(D) constant

27. Inverse domain equals original
- **(A)** domain
- **(B)** range
- **(C)** x-intercept
- **(D)** slope

28. Inverse of $2x+3$ is
- **(A)** $(x-3)/2$
- **(B)** $(x+3)/2$
- **(C)** $2x-3$
- **(D)** $1/(2x+3)$

29. Horizontal line test checks
- **(A)** function property
- **(B)** one-to-one property
- **(C)** continuity
- **(D)** asymptote

30. f^{-1} means
- **(A)** reciprocal
- **(B)** inverse function
- **(C)** negative function
- **(D)** derivative

AP-Style Free Response

31. Let $f(x)=\frac{2x+5}{x-1}$. (a) Find inverse. (b) State domains and ranges. (c) Verify one composition.

32. A temperature conversion is $F(C)=\frac{9}{5}C+32$. (a) Find inverse. (b) Interpret it. (c) A sensor only operates for -20<=C<=50; state inverse domain/range under restriction.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.